Field Service & Asset Maintenance Management

Angular supervisor dashboard / ASP.NET Core 10 / .NET MAUI technician application / PostgreSQL and SQLite

System overview

This project implements a local asset-maintenance workflow: supervisors register sites and assets, generate work orders, assign technicians and review inspections. The Android-targeted .NET MAUI application records the technician's checklist, notes, faults, part quantities and photographs while offline. The authenticated ASP.NET Core API persists authoritative work orders and maintenance history in PostgreSQL.

Application responsibilities

Supervisor (Angular)	Create sites, assets, schedules, assignments and stock adjustments; review submitted work and read reports/history.
Technician (.NET MAUI)	Download assigned jobs, edit offline, cache photos, submit inspections and explicitly resolve conflicts.
Offline engine (SQLite)	Keep per-user drafts, immutable outbox requests, photo bytes, acknowledgments and archived snapshots across restarts.
API and PostgreSQL	Enforce roles and assignment, validate versions, transact inspection/stock/faults/audit/receipt together and record supervisor completion.

Offline delivery and concurrency

Before a network request, the mobile engine freezes the operation ID and exact payload. Retry and process restart reuse that same request, even when the server committed and its response was lost. The API stores the response receipt in the same serializable transaction as inspection changes, stock deductions, faults and audit. A repeated operation ID with different content is rejected. Uploaded photos must be acknowledged before an inspection refers to them.

The app distinguishes Pending, Syncing, Failed, Conflict and Synced from the server's maintenance status. A stale version or reassignment preserves local work. Conflict resolution is explicit: adopt the server or rebase an archived local draft, then submit again. Technicians cannot complete their own jobs; a supervisor reviews Submitted work before recording Completed.

Verification and boundaries

Evidence recorded in the repository on 3 October 2026. These results describe a local implementation, not a production deployment.

Recorded checks

13 C# tests	Password hashing, lifecycle rules, SQLite persistence, photo recovery, exact lost-ack replay, immutable requests and explicit conflict resolution.
Live API + PostgreSQL	Role enforcement, assignment checks, version conflicts, safe photos, exactly-once stock effects, scheduling, supervisor review, history and reporting.
Actual SQLite core + live API	Offline inspection, process restart, network loss, a response deliberately lost after commit, preserved photo bytes, and correction after stock rejection.
Angular browser checks	Supervisor sign-in, persisted metrics, schedule/version handling, inspection evidence, stock adjustment and reports.
Android build	Debug APK compiled and exported with zero compiler warnings or errors. Native GUI/emulator acceptance was deferred.

Security and transaction boundaries

Supervisors and technicians authenticate with JWTs. The API checks roles, record ownership and assignment; technician mutations require the expected work-order version. Upload limits, image validation and required checklist checks are enforced server-side. Photo bytes and maintained records live in PostgreSQL, while the device isolates drafts by signed-in user. Local Android debug transport permits the emulator host only; production deployment requires trusted HTTPS and managed credentials.

Not verified or not included

The full Android native interface journey (camera/picker permissions, SecureStorage and page lifecycle) was not exercised on a device, despite a successful APK build. iOS and Mac Catalyst were not built or run. There is no public production deployment, refresh-token/password-reset flow, push dispatch, automatic schedule runner, or automatic per-field conflict merging. GitHub Actions workflows are configured, but remote execution is not claimed.

Repository and implementation

Source: github.com/keenosmith-del/field-service-and-asset-maintenance
Primary references: README.md, docs/verification.md, docs/requirements-signoff.md, docs/redesign-handoff.md.